Institute of Chemical Technology, Mumbai

Department of Mathematics

Machine Learning and Artificial Intelligence (MAP 1602)

Audio Signal Processing For Machine Learning

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Problem Statement

Aim

To understand how Audio and Text data is processed for Machine Learning.

Motivation

Unstructured Data: Working with Unstructured Data (Audio, Image, Text) can give competitive edge.

Domain-Specific Analysis: To understand concepts related to Acoustics, speech recognition, TTS and many more

Expected Outcome

Exploratory/Descriptive

To visualize audio data and extract meaningful features for analysis and observations

Data Source and Basic Description

1. Overview

The dataset consists of approximately **10 hours of standard studio-quality Hindi speech recordings** produced by a **single female speaker**, along with corresponding Hindi text transcripts.
This dataset is well-suited for **Text-to-Speech (TTS)** research and audio signal processing tasks and represents a **multimedia dataset**.

Dataset Source (AI-Kosh):
https://aikosh.indiaai.gov.in/web/datasets/details/hindi_female_mono_indictts_phase3.html

2. Dataset Composition and Statistics

Total Audio Files: 4,586
Total Duration: Approximately 10 hours
Language: Standard Hindi
Speaker Profile: Single Female, Monolingual
Recording Environment: Standard studio-quality recordings with minimal background noise

3. Technical Specifications (Audio)

File Format: Waveform Audio File Format (.wav)
Sampling Rate: 48,000 Hz (48 kHz)
Bit Rate: 768 kbps
Audio Channels: Mono (1 channel)
Bit Depth: 16-bit

4. Dataset Structure and Annotation

The dataset is organized into two primary components to enable accurate alignment between audio signals and their corresponding textual content:

Hindi fem audio (Audio Repository):
Contains 4,586 individual .wav audio files.
Each file is uniquely named (e.g., train_hindifem_00001.wav, train_hindifem_00002.wav) for systematic indexing.
Hindi fem mono.txt (Transcript File):
Serves as the **ground-truth text annotation file**.
Provides a **one-to-one mapping** between each audio file and its corresponding Hindi transcript.

Format Example:

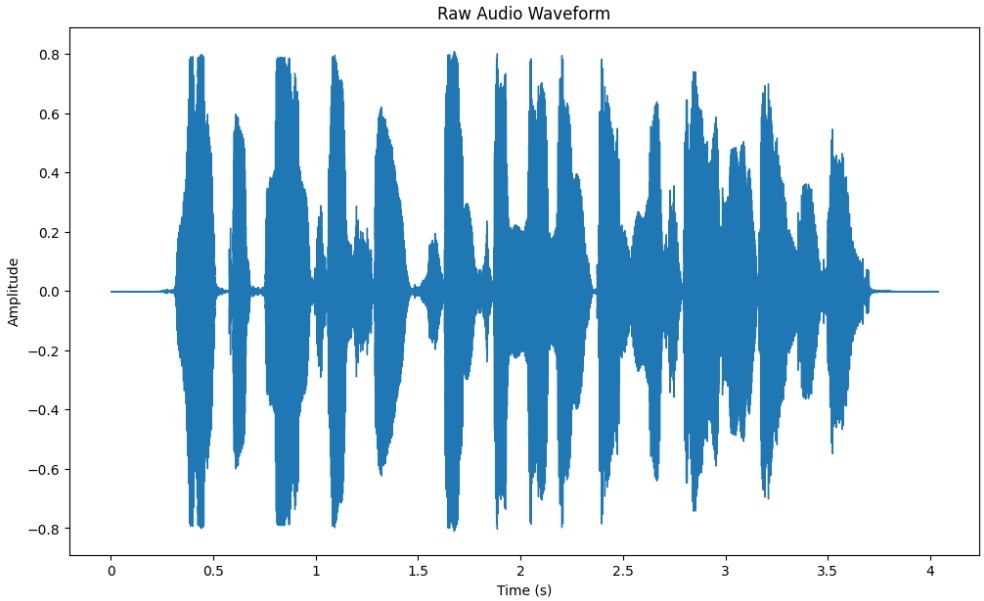
train_hindifem_00001.txt मैं जीवन की राह में आगे बढ़ चला ।

strain_hindifem_00002.txt तारीख के दिन मेरे स्वामी जाएँगे और लेदेकर मामला रफ़ादफ़ा हो जाएगा

Exploratory Data Analysis

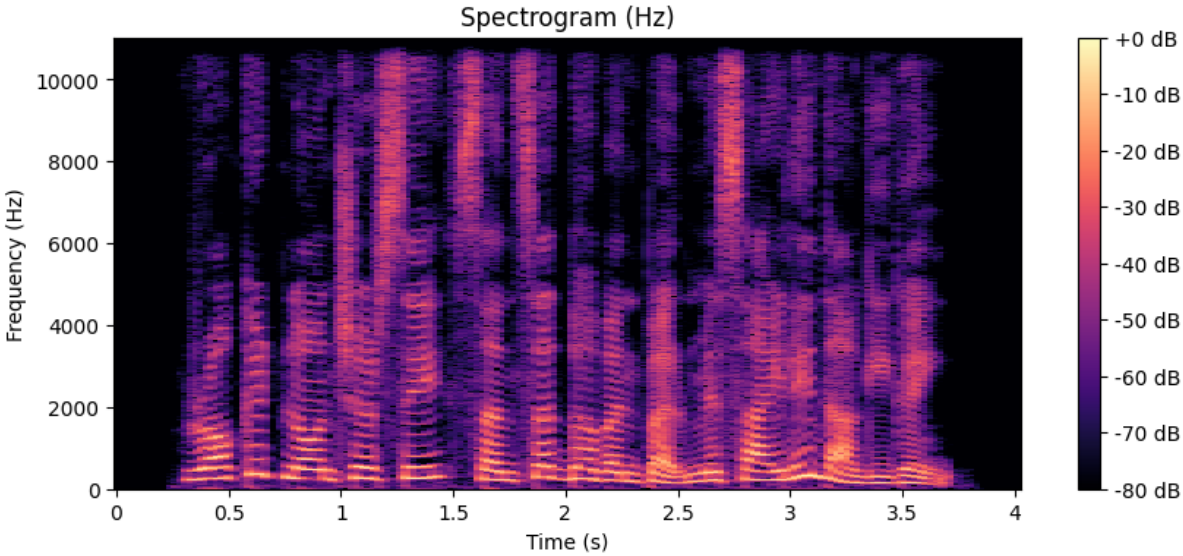
1. Visualizations

1. Raw Audio Waveform (Time Domain)



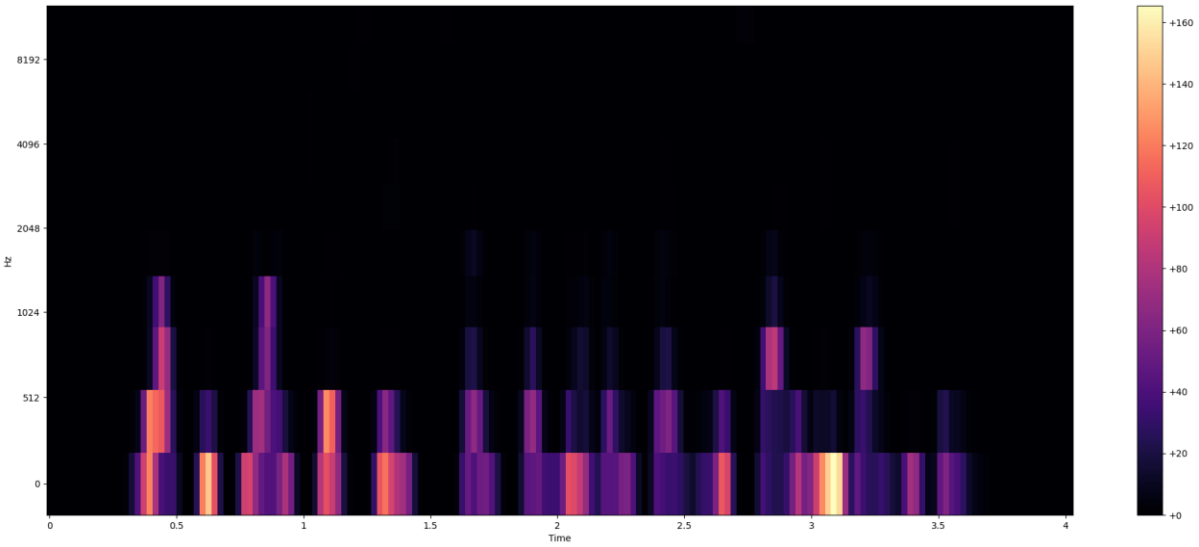
Represents the variation of the audio signal amplitude over time, helping identify loud and quiet regions in the recording

2. Spectrogram (Frequency Domain)



Illustrates how the frequency components of the audio signal change over time, with color intensity indicating signal strength. (We Use Short Time Fourier Transform to go form time domain to frequency domain)

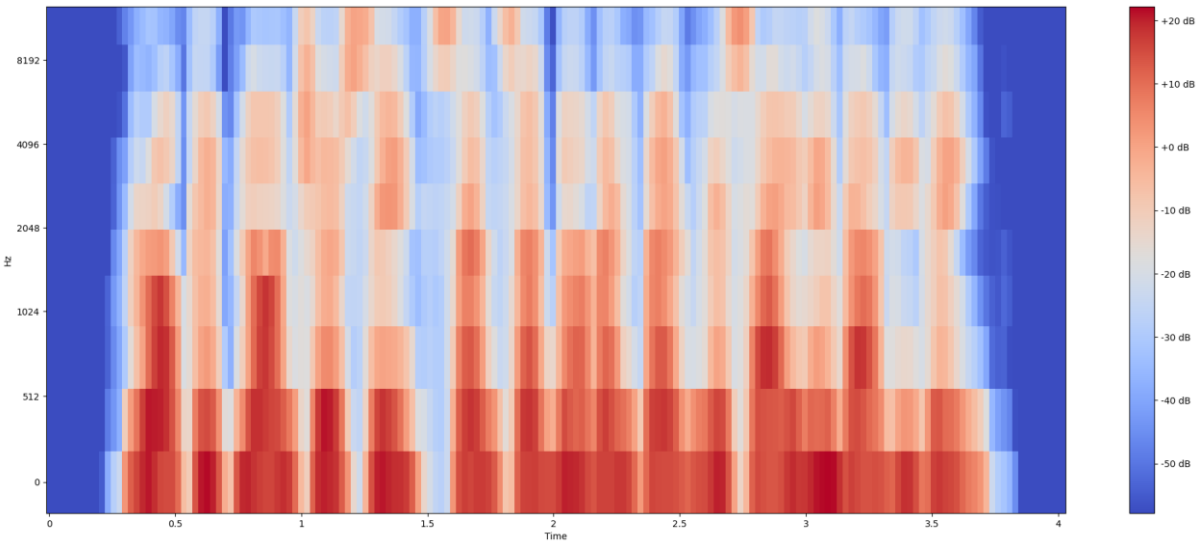
3. Mel Spectrogram



Mel Scale

The Mel scale relates the perceived pitch of a sound to its actual frequency. Humans can detect smaller pitch changes at low frequencies than at high frequencies, so using the Mel scale helps features better match human hearing.

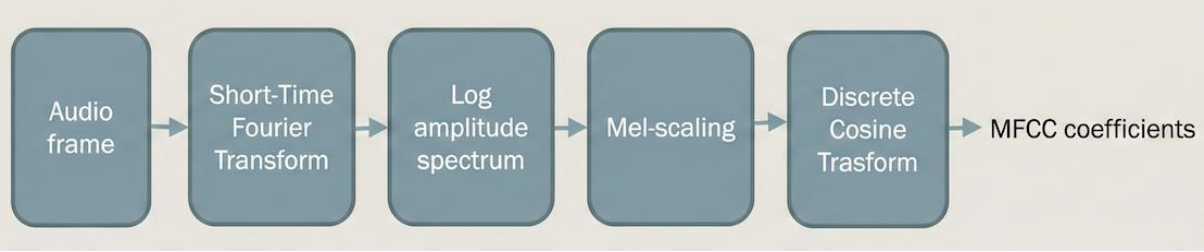
4. Log Mel Spectrogram



Represents the Mel spectrogram on a logarithmic scale, which highlights important sound patterns while reducing the effect of very large amplitude variations.

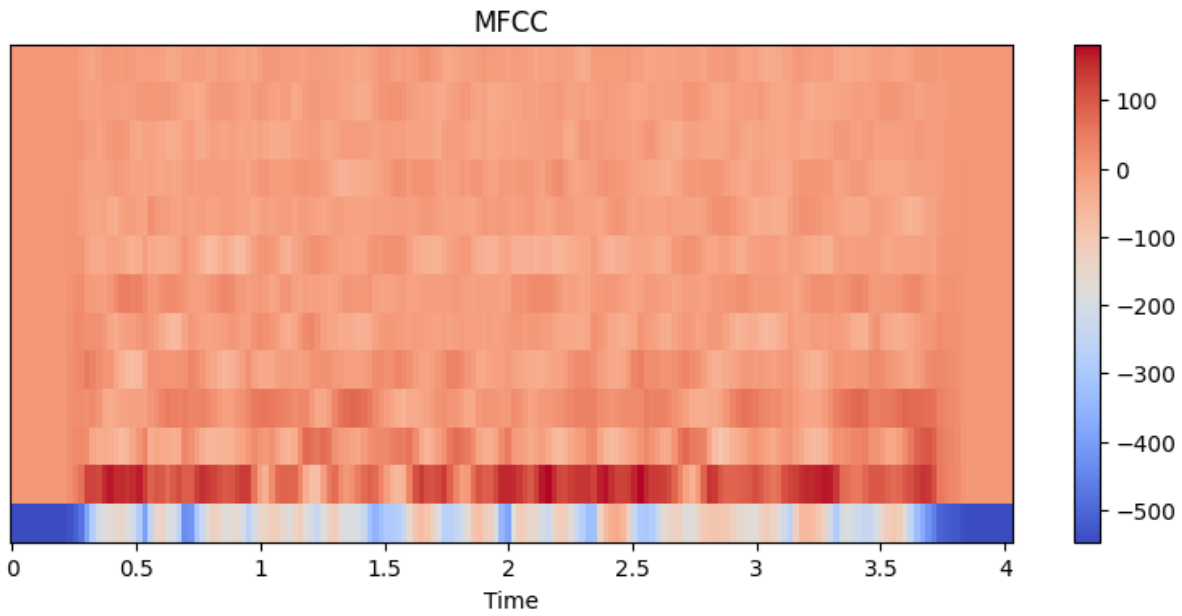
2. Feature Extraction

Every audio signal consists of many features. However, we must extract the characteristics that are relevant to the problem we are trying to solve.



In speech, the raw air from the lungs is filtered by the shape of the mouth, tongue, and throat (the vocal tract). MFCCs accurately capture this "shape" (timbre), making them ideal for identifying *what* is being said or *who* is saying it.

Mel-Frequency Cepstral Coefficients (MFCCs)

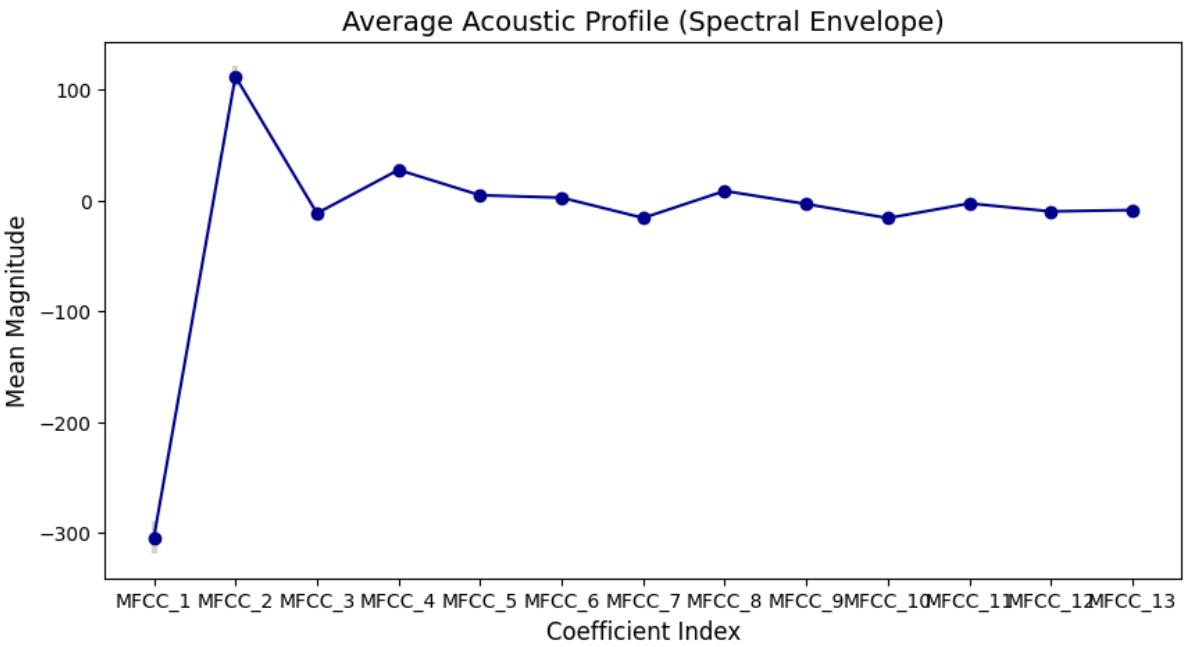


Mel-Frequency Cepstral Coefficients (MFCCs) reflect how humans perceive sound rather than how sound is physically recorded.

In practice, MFCCs are represented by a small set of coefficients. The **first 13 MFCCs** are the most commonly used because they capture the overall shape of the speech spectrum effectively.

- **MFCC 1** represents the overall energy of the signal.
- **MFCCs 2–13** describe the spectral shape and formant structure of the sound, which are crucial for distinguishing different speech sounds.

These 13 coefficients provide a compact yet meaningful representation of speech, making MFCCs highly suitable for speech recognition, speaker identification, and audio classification tasks



Standard Deviation

mfcc_1	14.629876
mfcc_2	9.457893
mfcc_3	5.875123
mfcc_4	6.478288
mfcc_5	4.950383
mfcc_6	3.816911
mfcc_7	3.817560
mfcc_8	3.614446
mfcc_9	2.825189
mfcc_10	3.448837
mfcc_11	3.591640
mfcc_12	2.419953
mfcc_13	2.517769

Mean

mfcc_1	-304.198921
mfcc_2	112.026276
mfcc_3	-11.472608
mfcc_4	27.720601
mfcc_5	4.816026
mfcc_6	2.630708
mfcc_7	-15.656410
mfcc_8	8.708613
mfcc_9	-3.109346
mfcc_10	-15.834548
mfcc_11	-2.584724
mfcc_12	-9.766752
mfcc_13	-8.544224

Observations

The minimal standard deviation across 4,586 samples validates the Studio Quality of the AI-Kosh dataset and confirms that the speaker maintained a consistent '**neutral**' tone throughout the recording process.

For this female speaker, a high positive value of mfcc_2 indicates a **"Bright" or "Clear" voice**, where energy is concentrated in frequencies humans perceive as sharp and distinct.

Google Collab Link:

https://colab.research.google.com/drive/1J87knz6K6A3BAdgJVDsA0R3O1A4UEozB?usp=sharing

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